WORKED SOLUTION

STEP III 2011, Question 11

Mechanics • Rigid Body Rotation

STEP III 2011, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q11

The key idea

After the twist, every string still runs from its fixed ceiling point to the point of the rim that has moved round by θ. The horizontal part of each string is therefore a *chord* of the circle, so the geometry comes from looking down from above. By symmetry every string behaves the same way, so the vertical equilibrium fixes the tension, and the moments of the horizontal parts of the tensions about the axis give the couple. Once the disc is released, energy conservation does the rest.

Solution

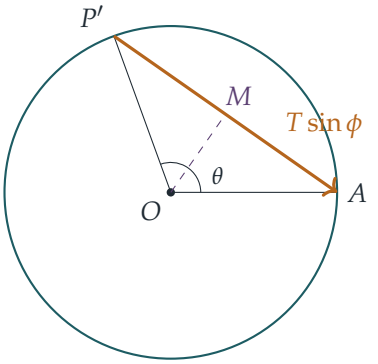
We model the disc as a rigid uniform lamina and the strings as light and inextensible. By symmetry, all the strings are taut with the same tension T and the same inclination ϕ, and the disc stays horizontal with its centre on the z-axis. When the disc is turned through θ, it rises to a depth h below the ceiling.

The angle of the string

The string through P is attached to the ceiling at $A = (a, 0, 0)$. After the rotation its lower end is at

$$P' = (a\cos\theta,\ a\sin\theta,\ -h).$$

Seen from above, A and P' both lie on the circle of radius a centred on the axis, with angle θ between them at the centre O. So the horizontal projection of the string is a chord of length $2a\sin\frac{1}{2}\theta$.



The view from above. M is the midpoint of the chord $P'A$, so OM bisects the angle θ, is perpendicular to the chord and has length $OM = a\cos\frac{1}{2}\theta$. The string itself has length b and makes angle ϕ with the vertical, so its horizontal projection has length $b\sin\phi$. Equating the two,

$$b\sin\phi = 2a\sin\tfrac{1}{2}\theta,$$

as required. Its vertical projection gives the new depth of the disc:

$$h = b\cos\phi = \sqrt{b^2 - 4a^2\sin^2\tfrac{1}{2}\theta}.$$

This is real because $b > 2a$. Since $0 < \theta < \pi$, we have $0 < \sin\frac{1}{2}\theta < 1$, so $0 < \phi < \frac{1}{2}\pi$.

The couple

Resolving vertically for the disc, the n equal tensions support the weight:

$$nT\cos\phi = mg.$$

The couple C acts in the plane of the disc, so it has no effect on this balance.

Each tension has a vertical component and a horizontal component $T\sin\phi$ along the chord from P' towards A. The vertical components are parallel to the axis, so they have no moment about it. The horizontal component acts along the chord, whose perpendicular distance from O is $OM = a\cos\frac{1}{2}\theta$. Every string's horizontal pull tends to turn the disc back towards $\theta = 0$, so all n moments have the same sense. Taking moments about the axis,

$$\begin{aligned} C &= n\,T\sin\phi\cdot a\cos\tfrac{1}{2}\theta = mg\tan\phi\; a\cos\tfrac{1}{2}\theta \\ &= mg\,a\cos\tfrac{1}{2}\theta\cdot\frac{2a\sin\frac{1}{2}\theta}{\sqrt{b^2-4a^2\sin^2\frac{1}{2}\theta}} = \frac{mga^2\sin\theta}{\sqrt{b^2-4a^2\sin^2\frac{1}{2}\theta}}, \end{aligned}$$

using $nT = mg/\cos\phi$, then $\tan\phi = b\sin\phi/(b\cos\phi)$, and finally $2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta = \sin\theta$. This is the required magnitude.

Release from rest

Once the couple is removed, the disc untwists and falls. The strings are inextensible and stay taut, so they do no work: each string's tension is perpendicular to the velocity of its attachment point relative to the fixed ceiling point. Gravity is the only force that does work, so mechanical energy is conserved.

By the geometry above, when the disc has turned through α from the vertical-string position, its depth is

$$z(\alpha) = \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\alpha} = \sqrt{b^2 - 2a^2(1 - \cos\alpha)}.$$

The disc's kinetic energy is $\frac{1}{2}m\dot{z}^2 + \frac{1}{2}I\omega^2$, with $I = \frac{1}{2}ma^2$ about the axis.

Ask yourself When the strings are vertical, does the centre of the disc still have a vertical velocity?

No. Differentiating,

$$\dot{z} = \frac{\mathrm{d}z}{\mathrm{d}\alpha}\,\dot{\alpha} = -\frac{a^2 \sin\alpha}{z}\,\dot{\alpha},$$

which is zero at $\alpha = 0$. The strings are vertical at that instant, and the disc is at its lowest point, so all the kinetic energy is rotational there.

The disc falls from depth $\sqrt{b^2 - 4a^2 \sin^2 \frac{1}{2}\theta}$ to depth b. Energy conservation gives

$$\tfrac{1}{2} \cdot \tfrac{1}{2}ma^2\,\omega^2 = mg\left(b - \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\theta}\right),$$

that is,

$$\frac{a^2\omega^2}{4g} = b - \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\theta},$$

as required.

Checks

- *Virtual work.* Turning the disc through a further $\delta\theta$ against the couple raises it by $-\delta z$, so $C = mg\,|\mathrm{d}z/\mathrm{d}\theta| = mg\,a^2 \sin\theta / z$. This is exactly the couple we found by taking moments.
- *Small twists.* For small θ, $C \approx mga^2\theta/b$, which is the familiar restoring couple of a bifilar (here n-filar) suspension.
- *Numerical.* We placed the n strings explicitly in 3D and summed the moments of the tensions for random a, b, θ, m, n. The result agreed with the formula. We also integrated the motion numerically after release with $a = 1$, $b = 3$, $\theta = 2$, $g = 9.81$. That gave $a^2\omega^2/(4g) = 0.51651$ when the strings became vertical, matching $3 - \sqrt{9 - 4\sin^2 1} = 0.51651$.

Common slips

- Taking the lever arm of the horizontal tension as a rather than $a\cos\frac{1}{2}\theta$. The force acts along the chord, not along a tangent.
- Forgetting that the disc *rises* when it is twisted. This change of height is exactly where the energy comes from in the last part.
- Including the translational kinetic energy $\frac{1}{2}m\dot{z}^2$ without noticing that it vanishes when the strings are vertical.

Beyond STEP

This is a system with one degree of freedom whose motion is constrained. The depth is a fixed function of the twist angle, so the energy depends only on α and $\dot{\alpha}$. Lagrangian mechanics turns that observation into equations of motion. It also explains why the couple is just $mg\,\mathrm{d}z/\mathrm{d}\alpha$, which is the principle of virtual work. Linearising about $\alpha = 0$ gives simple harmonic torsional oscillations, which is how a bifilar suspension is used to measure moments of inertia. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation and energy methods at this level.

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